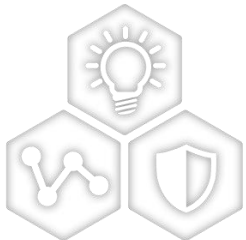


AI-powered Simulation Model Creation



A Leading Provider of Smart, Connected and Secure Embedded Control Solutions



SMART | CONNECTED | SECURE

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ATSO 2026-03-16

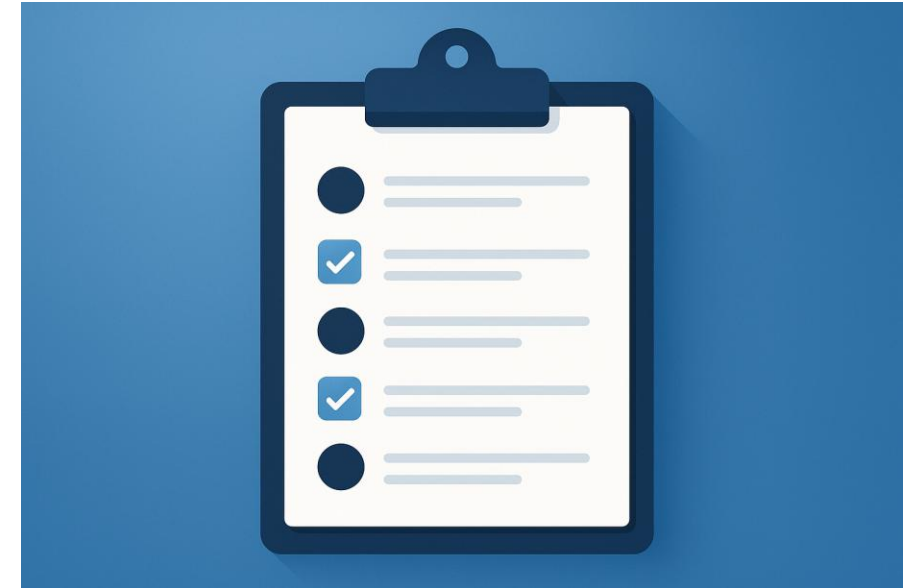
Executive Summary

Due to BU priorities and bandwidth, generic SPICE models are not available for most of MCHP's analog products.

In this presentation, the analog CAEs propose experimenting with a new process that uses generative AI to create SPICE models for customers on demand, solving the business and technical pain points.

Agenda

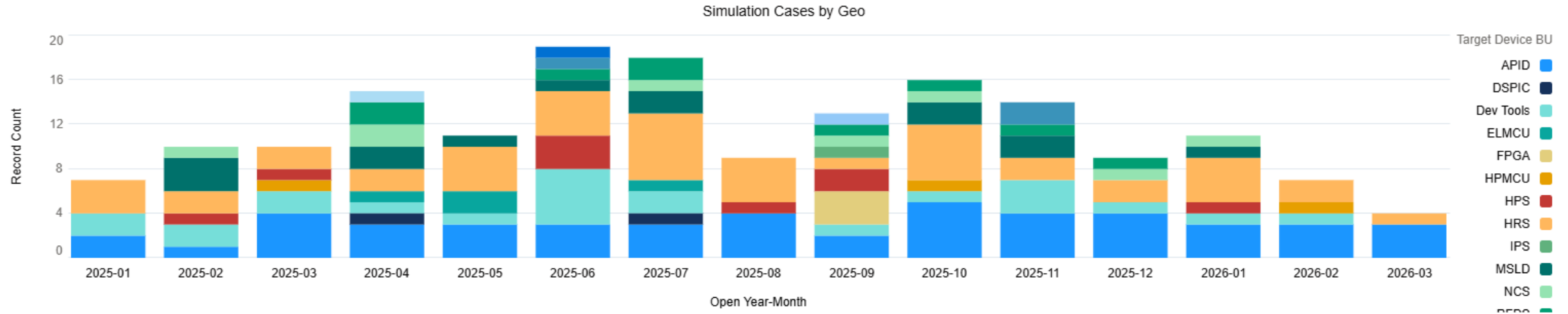
- **The Problem**
- **Actions So Far**
 - Discussion with APID
 - Proposal overview
- **Proposed Workflow**
 - Current process
 - New proposed Process flowchart
- **Preliminary Results**
 - Model generation
 - Model verification



The Problem

The Problem

Customer SPICE Model Requests



- Sometimes as many as 20 WW requests per month for SPICE models
- SPICE models generally not available
 - Even Mindi models often not available
- [Link to Salesforce report](#)

The Problem

Current Workflow is Inefficient

- **Model requests currently require a NSCAR**
 - NSCAR process is long and involves many stakeholders
 - MCHP must have high certainty of good ROI
- **Model creation is expensive and time-consuming**
 - Outsourcing creates communication inefficiency
 - Vendor: External development (OrCAD/Cadence) at \$178/hr; costs can reach \$20K.
- **BUs have different priorities and are bandwidth limited**
 - Focus is usually on high-volume or new products
 - Does not make business sense to fill in back catalog



Actions So Far

Actions So Far

Discussion with APID

- **Bi-monthly meetings with APID**
- **APID Design Engineering Team in Romania and CAE meeting**
- **Agreement between APID and CAEs to create a test model for MIC2775 (Reset supervisor)**



Proposal

- **Strategic Shift: AI vs. Outsourcing**

- Efficiency
- Time Savings
- Cost Optimization

- **Pilot Program & Execution**

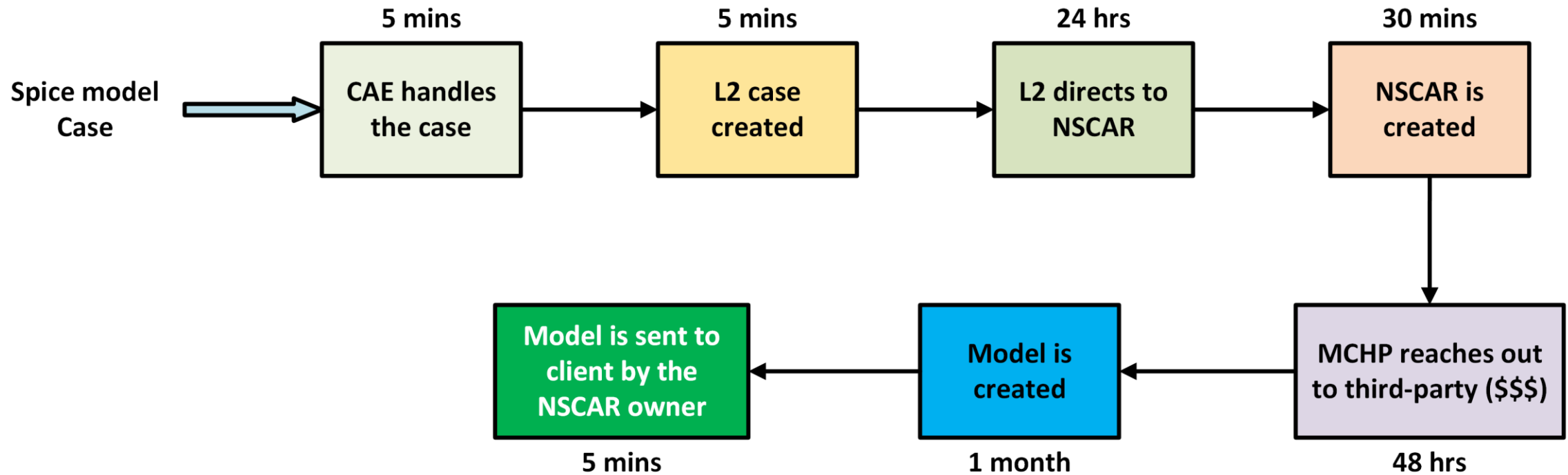
- High-Demand Candidates: Pilot evaluation will focus on APID components, specifically starting with MIC2775 (Reset Supervisor).
- CAE Integration: Leverage CAE resources to drive internal development and support

- **Validation**

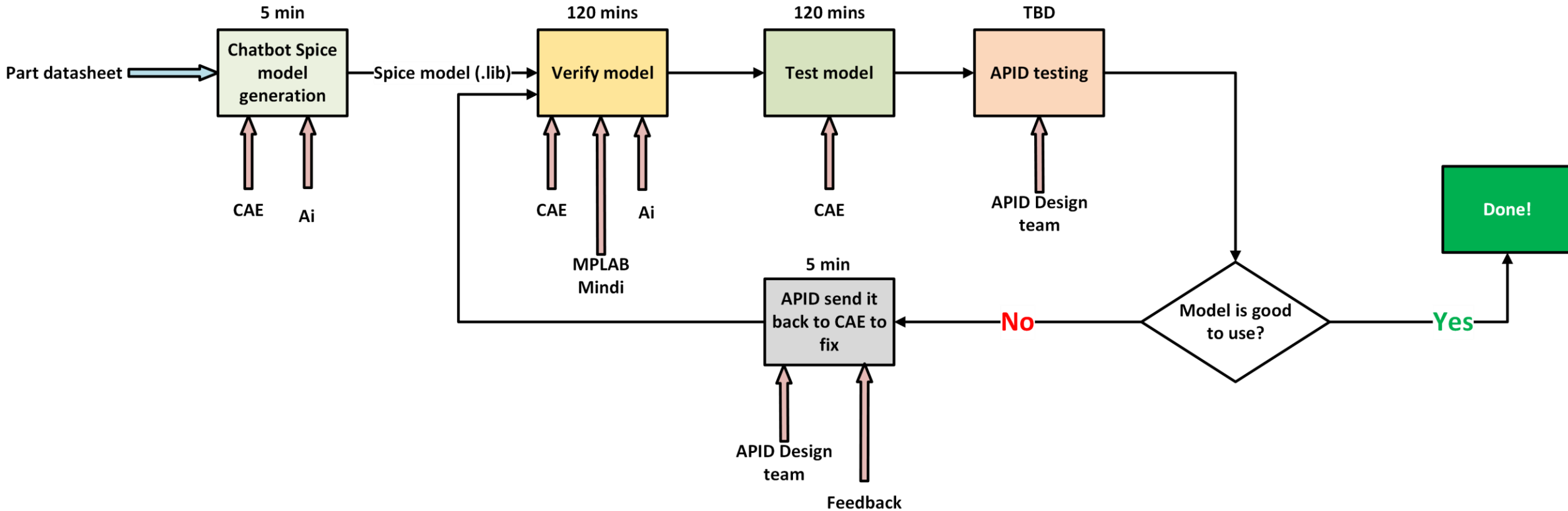
- Establish a dedicated group consisting of one BU Applications Reviewer and interested CAEs
- Ensure all AI-generated models undergo rigorous validation to match physical product performance
- Challenge: Validation bandwidth is a bottleneck due to competing NPI priorities

Proposed Workflow

Current Workflow



New Process Workflow



Compare and Contrast

	Current Process	Spice model AI creation
Time	~1+ month	~2 weeks
Effort	• CAE • BU Apps • NSCAR team • Third-party vendor	• CAE • BU Apps
Budget	~\$20,000	Included in MCHP Opex

Preliminary Results

Preliminary Results

First attempts at chatbot SPICE model generation

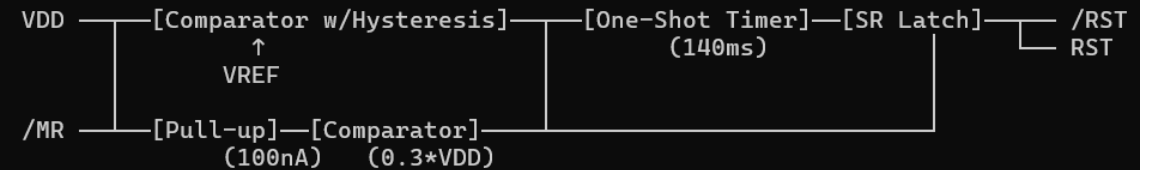
- Simple MCHP Chatbot and Gemini prompts produced unusable models
- A few iterations using Claude Code produced the first usable model
 - Some behaviors of the model were incorrect but more iterations solved this

• I've created a behavioral SPICE model for the MIC2775. Here's a summary:

Files Created

File	Description
MIC2775.lib	Main SPICE model library
MIC2775_test.cir	Test bench for verification

Model Architecture (Based on Block Diagram)



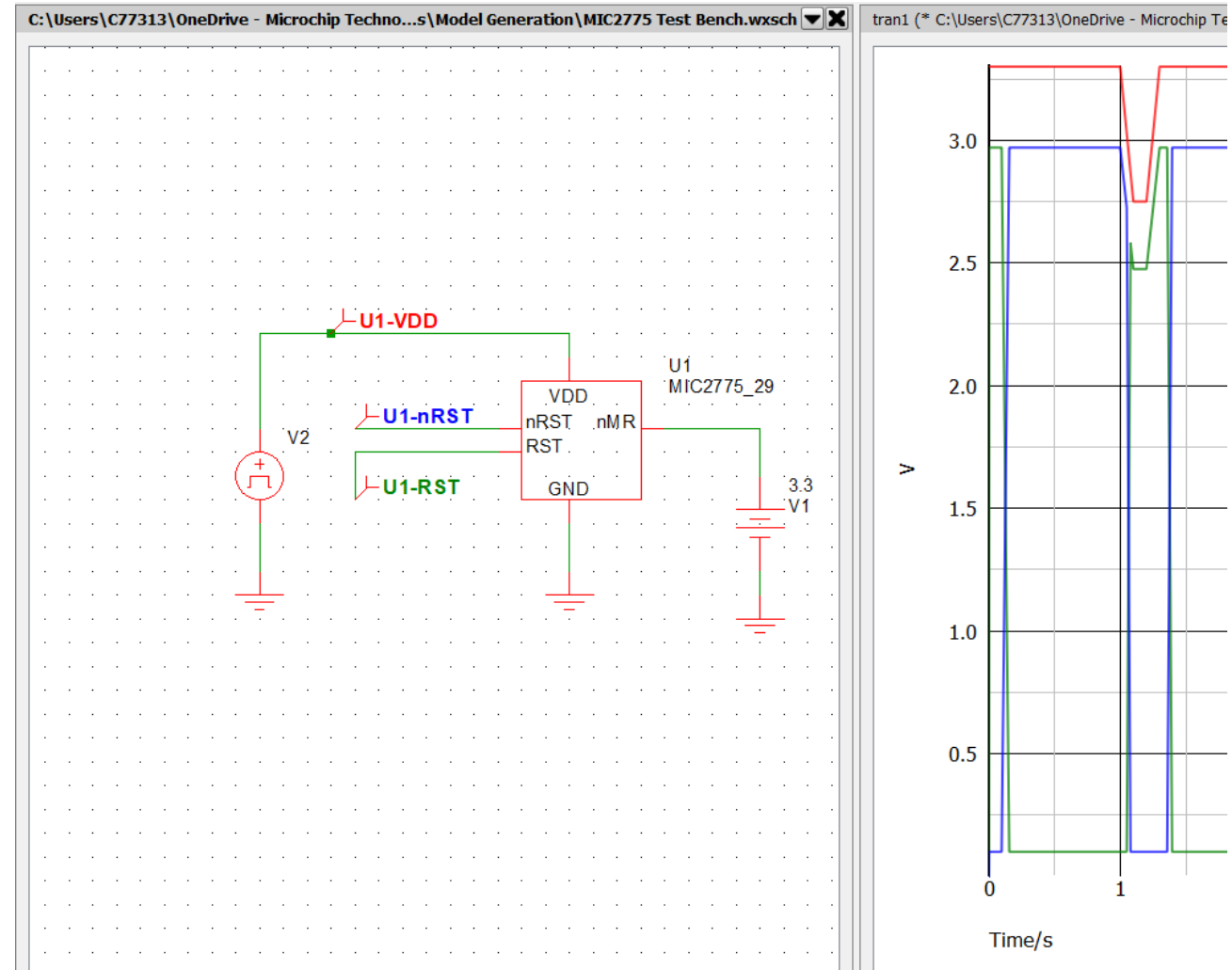
Key Parameters from Datasheet

Parameter	Value	Implementation
Threshold (VTH)	1.69V - 4.68V	Selectable via variant
Hysteresis	1% of VTH	Built into comparator
Reset Pulse Width	140ms min	One-shot timer
Propagation Delay	20μs (comp), 5μs (/MR)	RC delay networks
/MR Pull-up	100nA	Current source
VOL	0.3V max	Output driver

Preliminary Results

Using Mindi to test model behavior

- Mindi doesn't accept all SPICE syntax
 - SPICE doesn't seem to be a well-unified standard
- Eventually we hope to automate the model evaluation
 - Use APID model verification process



Questions



Thank you!
